Progress Report: Curried Functional Spaces

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Due date: Fri, Mar 27
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1. Abstract

The study of real-valued multivariable functions is traditionally done by expanding the domain from $\mathbb{R}$ to $\mathbb{R}^n$. However, there is an alternative way to represent a function of many arguments, primarily used in a field of mathematical logic called *lambda calculus.*[1] This involves treating a function $f : \mathbb{R}^n \to \mathbb{R}$ as a single-variable function that receives a real number $x_1 \in \mathbb{R}$ and produces *another function* $f_{x_1} : \mathbb{R}^{n-1} \to \mathbb{R}$, now of $n - 1$ arguments. In other words, we are "partially applying" the function f to the argument x_1. Continuing this process, we obtain what is called the *curry* of the function f, denoted $\gamma(f)$:

$$\gamma(f) : \mathbb{R} \to (\mathbb{R} \to (\mathbb{R} \to \ldots \to (\mathbb{R} \to \mathbb{R})\ldots))$$

Both these representations are equivalent in the sense that the correspondence $f \leftrightarrow \gamma(f)$ constitutes a bijective map. A natural question arising from this observation, is whether this equivalence stretches over continuity and differentiability, and whether one can obtain non-trivial results about a multivariable function by studying its curry.

2. Survey of the literature

The study of curried functions naturally brings us to the field of functional analysis, since it involves inductively building spaces of functions whose codomain is again a space of functions. A number of fundamental works are instrumental here:

(1) N. Bourbaki, *General Topology, Part II,* 1966. This textbook deals (among all else) with topologies on various sets of functions $f : X \to Y$, where X and Y are topological spaces. Of particular interest are the *topology of pointwise convergence* which facilitates the general curried function space $H_n(\mathbb{R})$, and the *compact-open topology* which serves as the basis for the spaces $C_n(\mathbb{R})$ of continuous curried functions.

(2) W. Rudin, *Functional Analysis,* 1991. This work introduces the theory of topological vector spaces which are central in functional analysis. In particular, it builds the toolkit for effectively working with neighborhoods of zero, and also provides more advanced results concerning metrizability and completeness.

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Bibliography

1. Barendregt H. *The Lambda Calculus, Its Syntax and Semantics.*; 1984.